

Mushfiqur Rahman Chowdhury

PhD Student in Computer Science | New Jersey Institute of Technology

Email: mushfiqurchowdhury05@gmail.com | mc2399@njit.edu

Cellphone: +1201-726-5104

Portfolio: <https://mushfiqur-chowdhury05.github.io/>

EDUCATIONAL BACKGROUND

New Jersey Institute of Technology, New Jersey, USA

Aug 2026 - Present

PhD in Computer Science

- Advised by Dr. Martin Kellogg

University of Dhaka, Bangladesh

Jan 2019 - Dec 2023

BSc. in Software Engineering

- CGPA: 3.79/4.00
-

RESEARCH EXPERIENCE

Researcher/Incoming Research Assistant

May 2026 - Present

New Jersey Institute of Technology

- Collaborating with Dr. Martin Kellogg on automated type inference research since May 2026.
- Developing automated approaches for type inference using program analysis and LLMs.
- Awarded a Research Assistantship beginning September 2026.

Researcher

May 2025 - April 2026

University of Texas at Arlington, TX, USA

- Worked with Dr. Faysal Hossain Shezan to develop a methodology that would allow KLEE based symbolic execution on large scale tightly coupled projects. The work focused on applying the methodology and detecting vulnerabilities in the Tensorflow Github repository.
- **Technical Skills:** Program analysis, Automated Software Testing, LLMs for Software Engineering, C/C++

Researcher

Jun 2025 - Present

University of New Mexico, NM, USA

- Working under the supervision of Dr. Abdullah Mueen to develop a tool that would perform analysis on the Ethereum blockchain data to link address clusters based on temporal patterns.
 - **Technical Skills:** Blockchain, Data Mining, Python
-

RESEARCH PAPERS & ABSTRACTS

Mushfiqur Rahman Chowdhury. "Slice-Guided, Context-Augmented Large Language Model Inference of Java Nullability Annotations." SPLASH/ISSTA 2026 Student Research Competition (SRC), 2026. **Accepted.**

TEACHING EXPERIENCE

Lecturer Sep 2025 - Present (On study leave since Jul 2026)
Department of Computer Science and Engineering
Daffodil International University, Dhaka, Bangladesh
Conducted classes and evaluation, designed course materials and created assignments
Courses: Natural Language Processing, Introduction to Data Science, Compiler Design Lab, Computer Graphics
Credentials: <https://faculty.daffodilvarsity.edu.bd/profile/cse/mushfiqucse.html>

INDUSTRY EXPERIENCE

Software Engineer (Embedded Systems) Dec 2023 - Sep 2025
Samsung R&D Institute Bangladesh, Dhaka, Bangladesh

- **Samsung Buds Audio Project**
 - Developed and deployed new features on the Galaxy Buds platform, with a focus on Galaxy Buds 3 and Galaxy Buds 3 Pro devices.
 - Diagnosed and resolved over 100 bugs reported by the testers and market users by analysing the device logs from the earbuds, identifying the root cause of the bugs and developing patches to fix the bugs.
- **Research & Innovation**
 - Submitted DOI on preserving privacy for virtual assistants in multi-user scenario
 - Developed a PoC for determining the stress level of a user based on data collected from earbud sensors.
- **Technical Skills:** Real-time Operating Systems (RTOS), Android Studio, LLM security, Deep Learning, C/C++, Java, Kotlin

Experience Certificate: https://mushfiqu-chowdhury05.github.io/files/SamsungExp_MR_Chowdhury.pdf

Software Engineering Intern Jan 2023 - Jul 2023
Samsung R&D Institute Bangladesh, Dhaka, Bangladesh

- I developed a prototype using an STM32 microcontroller that imitates and demonstrates the use of schedulers in the Galaxy Buds platform.
- I developed an employee evaluation system that generate a quiz based on pre-defined questions and scores the employees of the company.
- I conducted multiple knowledge transfer sessions with my teammates.

RESEARCH PROJECTS

Slice-Directed Type Annotator May 2026 - Present
New Jersey Institute of Technology, NJ, USA

- Working with Dr. Martin Kellogg to develop an automated type inference tool that is independent of type systems.
- It uses the type-directed slicing tool Specimin, to generate program slices to reduce the type inference problem for a large project to a small subsystem and then applies LLM to infer types for a program repository.

ExtraKLEE**May 2025-April 2026****University of Texas at Arlington, TX, USA**

- Under the supervision of Dr. Faysal Hossain Shezan, I developed a tool prototype that would perform symbolic execution on any large scale code repository such as Tensorflow.
- The approach generates a test harness for testing particular Tensorflow APIs and extracts its related dependencies.
- **Github Link:** <https://github.com/mushfiqiit/ExtraKLEE>
- **Technical Skills:** C++, Program Analysis, Automated Software Testing.

BorderBlaze: Boundary Point Segmentation from Aerial LiDAR Point Cloud Data**University of Dhaka, Dhaka, Bangladesh****Jul 2023-Dec 2023**

- Developed a web-based tool to evaluate boundary point segmentation algorithms on a LiDAR point cloud dataset of building roof structures in a landscape.
- Implemented three different algorithms for evaluation: Neighbourhood Based Approach, Concave Hull Algorithm and Delaunay Triangulation.
- **Github Link:** <https://github.com/mushfiqiit/BorderBlazeSPL3>
- **Technical Skills:** FastAPI, Python, Computer Vision

Bangladesh Institute of Capital Market (BICM) website**Apr 2022-Nov 2022****Bangladesh Institute of Capital Market, Dhaka, Bangladesh**

- Developed the backend of the official BICM website that uses python web scraping techniques to extract stock market data from the official Dhaka Stock Exchange (DSE) website.
- Solved multiple vulnerabilities of the application by using context-based program analysis techniques which were exposed due to the dynamic changes in the DSE website.
- Website Link: <https://dfl.bicm.ac.bd/>
- **Technical Skills:** Python, Django, Web scraping, Program analysis.

ACADEMIC PROJECTS

Save The Date: A Party Planner**Jan 2022-Jun 2022****University of Dhaka, Dhaka, Bangladesh**

- Developed a party planning web application which serves as a centralised hub, connecting various party service providers (caterers, decorators, vendors etc.) with potential customers.
- Developed a recommendation system using K-Nearest Neighbor algorithm to generate a complete plan based on user inputs such as, date, budget, and location.
- **Github Link:** <https://github.com/saleheenshafig9/SaveTheDate>
- **Technical Skills:** Django, Python, ReactJS

ACHIEVEMENTS

Excellence Award, Samsung R&D Institute Bangladesh

Received excellence award in recognition of my contribution to the company during October-December of the year 2024. During that time, I developed a PoC for calculating stress score of an individual using earbud's sensor data.

Certificate: <https://mushfiqur-chowdhury05.github.io/files/ExcellenceAward.pdf>

International Collegiate Programming Contest (ICPC) Dhaka Regional Contest 2021

- **Official Rank:** 23 (qualified for the contest from a pool of 1747 teams participating across Bangladesh)
- **Team Name:** DU_Untitled(4)
- **Standings:** <https://icpc.global/regionals/finder/Dhaka-2022/standings>
- **ICPC ID:** <https://icpc.global/ICPCID/OTLKXQQ6H3EW>

International Collegiate Programming Contest (ICPC) Asia Dhaka Regional Site Online Preliminary Contest 2020

- Rank: 49 among 1272 teams
- Team Name: IITDU_DeathStars
- Standings: <https://icpc.global/regionals/finder/Dhaka-Preliminary-2021/standings>

SKILLS

Programming Languages: Python, C/C++, Java, JavaScript, PHP, Kotlin

Frameworks/Libraries: FastAPI, Django, ReactJS, MySQL, Real-time OS, Android SDK

Soft Skills: Planning, Presentation, Teamwork, Documentation, Leadership

REFERENCES

Martin Kellogg
Assistant Professor
New Jersey Institute of Technology
NJ, USA
Email: martin.kellogg@njit.edu

Abdullah Mueen
Professor
University of New Mexico,
NM, USA
Email: mueen@unm.edu